

Who Does My Child Believe: Me or AI?

A home-based event-boundary paradigm for epistemic deference (trust) in early childhood

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19 **Abstract**

20 Children are increasingly exposed to generative artificial intelligence (AI) systems during critical
21 periods of cognitive development, yet the effects of such exposure on selective trust remain
22 poorly understood. This study introduces a home-based event-boundary paradigm designed to
23 test epistemic deference within a child–caregiver–AI triadic context. One child (aged 3 years)
24 and a caregiver viewed a 2:04-minute clip from *Bluey* with four predefined pauses: two at event
25 boundaries (high narrative uncertainty) and two at control moments (low uncertainty). At each
26 pause, the child completed three tasks: event segmentation (“same/new”), narrative prediction,
27 and deference choice under caregiver–AI disagreement. Informant accuracy was counterbalanced
28 by design (AI correct at boundaries; caregiver correct at controls; both 2/4 overall), isolating
29 contextual uncertainty from cumulative reliability learning. As a proof-of-concept pilot ($N = 1$),
30 the study targeted feasibility and mechanism-level hypothesis refinement rather than population-
31 level effect estimation. The protocol was fully executable and produced complete trial-level data.
32 Overall robot choice was 3/4, yet robot deference was lower at boundaries (.50) than controls
33 (1.00), yielding a negative boundary deference effect ($\Delta = -.50$). Segmentation and prediction
34 accuracy dropped from perfect in the first half (1.00) to null in the second half (.00), suggesting
35 attentional or fatigue contamination. The pilot does not confirm the original directional
36 hypothesis; instead, it supports a refined model in which epistemic deference reflects interactions
37 among novelty bias, recency-weighted reliability tracking, and subjective boundary detection.
38 These findings justify progression to a preregistered, adequately powered confirmatory study.

39 **Keywords:** selective trust, epistemic deference, event segmentation, artificial intelligence,
40 cognitive offloading, early childhood, proof-of-concept pilot

41 1. Introduction

42 From at least the age of three, children are not indiscriminate consumers of testimony.
43 They express trust by integrating informant characteristics (e.g. familiarity, past accuracy, and
44 attributed expertise) into their learning decisions (Harris & Corriveau, 2011; Koenig & Harris,
45 2005). This capacity for *selective trust* is a cornerstone of cultural learning. It enables children to
46 acquire knowledge efficiently while filtering unreliable sources (Tomasello et al., 2005).
47 Crucially, the relation between an informant's accuracy history and children's trust is robust
48 enough that it resists moderation by inferences about the informant's traits or intentions (Ronfard
49 & Lane, 2018). Yet contemporary digital ecologies introduce a radically new epistemic actor into
50 the child's social world: generative artificial intelligence.

51 AI systems such as voice assistants, smart toys, and large language model interfaces are
52 now present in many households with young children (Xu, 2024). These systems generate
53 predictions, offer explanations, and answer questions with surface fluency that can be mistaken
54 for genuine competence, a phenomenon that recent work has termed the *confidence heuristic*
55 (Jose & Thomas, 2025). Emerging research on epistemic trust in generative AI demonstrates that
56 users, including students, often accept AI-generated output uncritically, engaging in what has
57 been called *automation bias* (Jose, B. et al., 2025). A growing body of work warns that heavy
58 cognitive offloading onto AI tools may produce *illusions of understanding* and reduce the kind of
59 effortful engagement on which deep learning depends (Fan et al., 2025; Gerlich, 2025). Yet
60 nearly all of this evidence comes from adults or adolescents. The developmental question, how
61 preschool children, whose epistemic vigilance mechanisms are still maturing (Sperber et al.,
62 2010), navigate trust decisions involving AI; remains largely unaddressed.

Event-segmentation theory provides a tractable framework for studying the cognitive conditions under which trust decisions occur. According to this theory, people spontaneously parse continuous experience into discrete events, and narrative boundaries (points at which the situational model must be updated) generate transient increases in prediction error and uncertainty (Zacks et al., 2007). We hypothesized that these moments of heightened epistemic openness could function as natural modulators of deference. Specifically, if a child's internal model is disrupted at an event boundary, the child may be more susceptible to external guidance, including guidance from a novel artificial informant.

A central methodological challenge in testing this hypothesis is separating the effects of contextual uncertainty from simple reliability learning. If one informant is consistently more accurate across trials, any observed deference pattern may reflect cumulative reinforcement history rather than boundary-specific epistemic openness. To address this confound, the present paradigm counterbalances informant accuracy by condition: the AI is correct at event boundaries, the caregiver is correct at control moments, and both informants end with identical overall accuracy (2/4 each). This design isolates the contextual effect of boundary uncertainty from the accumulation of differential reliability.

The original directional hypothesis predicted higher AI deference at event boundaries. However, because the paradigm is novel and involves a demanding triadic interaction with a 3-year-old in a domestic setting, a proof-of-concept pilot was necessary before committing to a powered confirmatory study. Consistent with best practice in experimental design (Leon et al., 2011), the primary goals of this pilot were: (a) to evaluate operational feasibility, (b) to test whether the measurement system yields interpretable behavioral variance, and (c) to identify design-level threats and refine the theoretical model.

86 2. Method

87 The present study employed a Single-Case Experimental Design (SCED) to evaluate the
88 operational viability of a novel domestic paradigm designed to measure epistemic deference
89 (trust) in early childhood. This pilot phase focused on a single three-year-old participant in an
90 ecological home setting. The primary objective was not to confirm generalizable effects but to
91 refine the experimental protocol and identify qualitative markers of cognitive conflict that group
92 averages might otherwise obscure.

93 2.1 Design

94 This was a single-case proof-of-concept pilot using a within-session repeated-measures
95 structure (four trials: P1–P4). The inferential target was feasibility and mechanism discovery, not
96 population-level confirmation.

97 2.3 Participant and setting

98 One child (3 years, male) participated with a primary caregiver in a domestic setting. The
99 caregiver was present throughout. The protocol was administered via a portable projector and
100 the child responded using physical cards depicting the caregiver and a cartoon robot. Informed
101 consent was obtained from the caregiver. The study complied with the Declaration of Helsinki
102 and institutional ethical guidelines. The study protocol was reviewed and approved by the
103 Research Ethics Committee of the University of Granada (CEIH), approval number
104 4948/CEIH/2025

105 2.4 Materials and stimulus

106 The stimulus was a 2:04-minute clip from *Bluey* Season 1, Episode 1 (“Magic
107 Xylophone”). *Bluey* was selected for its high narrative density, child appeal, and well-defined
108 event structure. Four pause points were pre-specified based on prior event-segmentation coding
109 of the episode:

110 P1 (Boundary, ~0:25): Scene transition—AI-correct trial.

111 P2 (Control, ~0:50): Within-scene continuation—caregiver-correct trial.

112 P3 (Boundary, ~1:15): Scene transition—AI-correct trial.

113 P4 (Control, ~1:35): Within-scene continuation—caregiver-correct trial.

114 2.5 Procedure

115 At each pause, the same three-step micro-protocol was administered:

116 *Segmentation*. The child was asked: “Is it the same thing happening, or is something new
117 happening?” This provides a behavioral index of subjective boundary detection.

118 *Prediction*. The child chose between two forced-choice options for what would happen next.
119 This assesses narrative prediction accuracy as a proxy for model-updating engagement.

120 *Deference*. The caregiver and the AI (represented by the robot card and a pre-recorded voice)
121 offered opposing predictions. The child selected which informant to follow. This is the primary
122 measure of epistemic deference.

123 After each response, playback resumed for approximately 10–15 seconds so the child
124 could observe the true outcome and update informant accuracy representations.

125 **2.6 Measures**

126 The primary construct was the *boundary deference effect*, operationalized as $\Delta_{\text{deference}} = P(\text{robot} |$
127 boundary) – $P(\text{robot} | \text{control})$. Secondary outcomes included, for example, segmentation
128 accuracy by condition and session half, prediction accuracy by condition and session half,
129 deference rate by condition, and hesitation (binary: present/absent) as a conflict marker.

130 **2.7 Analytic strategy**

131 Given $N = 1$ and four trials, analyses were descriptive and case-based: (a) trial-level
132 trajectory analysis, (b) condition-level proportions, (c) first-half versus second-half temporal
133 profile, and (d) exact binomial 95% confidence intervals for observed proportions. No
134 confirmatory population inferences were attempted.

135 **3. Results**

136 **3.1 Feasibility and data completeness**

137 The full protocol was completed (4/4 trials). All planned variables were captured without
138 missing values. The counterbalanced accuracy manipulation was successfully implemented as
139 designed. Administration time was approximately five minutes, well within the attentional
140 tolerance window for this age group.

3.2 Trial-level pattern

The deference sequence across trials was: Robot (P1), Robot (P2), Caregiver (P3), Robot (P4). Hesitation—operationalized as a visible delay or gaze alternation between response cards—occurred only at P1. This trial-level trajectory is displayed in Figure 1.

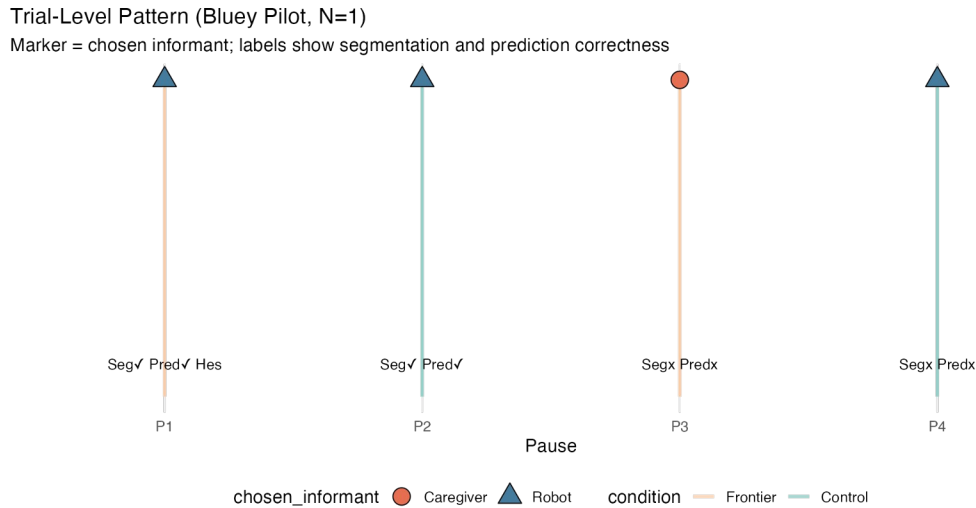


Figure 1. Trial-level response pattern across P1–P4. Markers indicate chosen informant (triangle = robot; circle = caregiver). Vertical bars are color-coded by condition (orange = boundary; teal = control). Bottom labels show segmentation and prediction correctness (✓ = correct; x = incorrect) and hesitation marker.

3.3 Condition-level outcomes

Overall robot choice was $3/4 = .75$, 95% CI [.194, .994]. Robot choice at event boundaries was $1/2 = .50$, 95% CI [.013, .987]; at control moments, $2/2 = 1.00$, 95% CI [.158, 1.000]. The boundary deference effect was $\Delta_{\text{deference}} = -.50$. These condition-level outcomes are summarized in Figure 2.

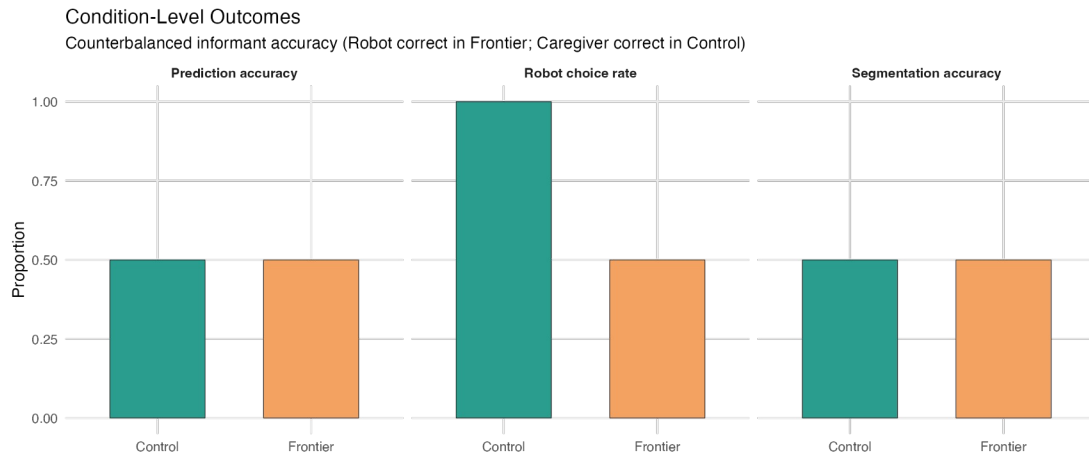


Figure 2. Condition-level outcomes for prediction accuracy, robot choice rate, and segmentation accuracy, disaggregated by condition (control vs. boundary). Informant accuracy was counterbalanced: robot correct at boundaries, caregiver correct at controls.

3.4 Temporal profile

Segmentation accuracy in the first half (P1–P2) was $2/2 = 1.00$, 95% CI [.158, 1.000]; in the second half (P3–P4), $0/2 = .00$, 95% CI [.000, .842]. Prediction accuracy followed an identical pattern: 1.00 in the first half, .00 in the second half. This complete first-to-second-half collapse, displayed in Figure 3, is consistent with attentional fatigue or disengagement in a four-pause sequence for this age group.

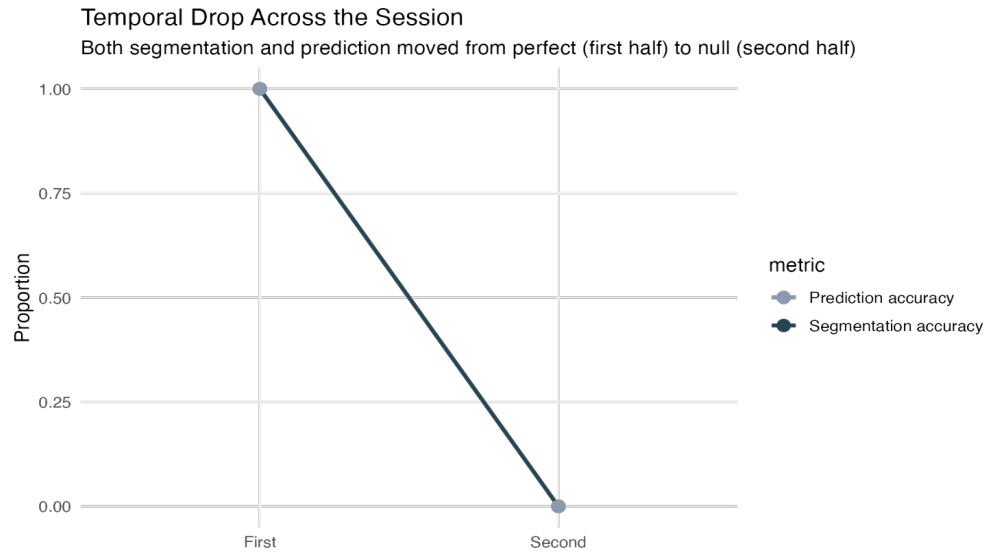


Figure 3. Temporal profile of segmentation accuracy and prediction accuracy by session half.

Both metrics dropped from perfect performance in the first half (P1–P2) to null performance in the second half (P3–P4), indicating attentional or fatigue contamination.

4. Discussion

4.1. Summary of findings

This pilot did not support the original directional hypothesis that AI deference would increase at event boundaries. The observed boundary deference effect was negative ($\Delta = -.50$), with the child choosing the robot *less* at boundaries than at control moments. Nevertheless, the pilot generated diagnostic evidence that is essential for theory refinement and design correction prior to a confirmatory study.

4.2. Mechanistic interpretation

Four patterns from the trial-level data warrant theoretical attention.

(1) *Baseline novelty attraction.* The child chose the robot in 3 of 4 trials, consistent with the well-documented novelty bias that preschoolers exhibit toward robotic and AI agents (Kahn et al., 2012). This baseline preference complicates the interpretation of any boundary-specific effect, because ceiling-level robot choice in one condition leaves little room for a directional contrast. A brief familiarization phase in the confirmatory design should attenuate this inflation.

(2) *Recency-weighted reliability tracking.* The single caregiver choice occurred at P3, immediately after the child had observed the robot give an incorrect prediction at P1 (visible after P1 outcome feedback). This is consistent with short-horizon reliability weighting: the child may have updated her informant model based on the most recent error rather than cumulative accuracy (Corriveau & Harris, 2009). Such recency effects are well-documented in the selective trust literature and pose a specific confound that the between-subjects confirmatory design must address.

(3) *Temporal degradation of cognitive engagement.* The abrupt collapse of both segmentation and prediction accuracy from perfect in the first half to null in the second half signals attentional fatigue. This is a serious validity threat: if the child is disengaged during P3–P4, deference choices in those trials cannot be meaningfully interpreted as reflecting epistemic processing. The confirmatory design should reduce to two pauses per child (one boundary, one control) to stay within the attentional envelope of 3-year-olds.

(4) *Epistemic conflict at first boundary.* Hesitation occurred uniquely at P1, the first boundary exposure. This singular hesitation event is consistent with heightened epistemic conflict—a moment in which the child’s internal model was genuinely disrupted, and the deference decision carried subjective weight. The absence of hesitation in subsequent trials may

200 reflect either resolution of the conflict or disengagement. In the confirmatory study, continuous
201 latency coding should replace binary hesitation to capture graded conflict.

202 4.3 Toward a Refined Model of Epistemic Deference

203 The original hypothesis assumed a simple main effect of event boundaries on AI
204 deference. The pilot data suggest that a more adequate model must incorporate at least three
205 interacting processes: (a) *novelty salience*—the baseline attractiveness of a novel artificial agent;
206 (b) *recency-weighted reliability memory*—the tendency to update informant trust based on the
207 most recent outcome rather than cumulative accuracy; and (c) *subjective boundary detection*—
208 the degree to which the child actually perceives a boundary, which may diverge from the
209 experimenter’s structural coding. This refined model aligns with recent accounts of AI as
210 epistemic infrastructure rather than mere tool (Chen, 2025), and with warnings about the erosion
211 of epistemic vigilance when children engage with confident-sounding artificial agents (Jose &
212 Thomas, 2025; Sperber et al., 2010).

213 4.4 Scientific validity of a single-case proof-of-concept

214 The validity claim of this study is narrow and explicit: it supports procedural and
215 diagnostic validity, not population validity. What is validly established includes: (a) *operational*
216 *validity*—the paradigm can be implemented end-to-end in a domestic setting with a 3-year-old;
217 (b) *measurement validity*—all three behavioral channels (segmentation, prediction, deference)
218 and the hesitation marker can be captured at trial resolution; (c) *manipulation validity*—the
219 counterbalanced informant-accuracy structure can be executed as designed; and (d) *mechanism*

220 *discovery value*—trial-level trajectories reveal concrete confounds (fatigue, novelty, recency)
221 that would remain invisible in aggregate-only reporting.

222 What is explicitly not claimed: no population prevalence estimate, no confirmatory test of
223 the original hypothesis, and no stable effect-size inference. This inferential boundary is
224 methodologically necessary. Proof-of-concept work answers a different question from
225 confirmatory work: “Does the design function, and which nuisance processes dominate?” versus
226 “What is the population-level effect under preregistered inference?” (Leon et al., 2011). The
227 present pilot answers the former and prepares the ground for the latter.

228 **4.5 Limitations**

229 The principal limitations of this pilot are threefold. First, the single-participant design
230 precludes any external generalizability. Second, the fixed within-session order induces
231 confounding by fatigue and recency that cannot be disentangled from condition effects. Third,
232 hesitation was coded as a binary variable; continuous response latency was not measured,
233 limiting sensitivity to graded epistemic conflict.

234 **4.6 Confirmatory next step**

235 A scaled confirmatory design should incorporate: (a) a between-subjects structure with
236 two pauses per child (one boundary, one control), eliminating within-session order and fatigue
237 confounds; (b) a brief pre-task robot familiarization phase to attenuate novelty inflation; (c)
238 continuous latency coding alongside binary deference choice as primary and secondary
239 outcomes; (d) preregistered hypotheses and a fully specified analytic plan deposited on OSF

240 prior to data collection; and (e) a target sample of approximately $N = 130$ (approximately 65 per
241 condition), based on power analysis for the primary binary endpoint.

242 **5. Conclusion**

243 As a proof-of-concept pilot, this $N = 1$ study is scientifically valid within explicit
244 inferential boundaries. Its contribution is not a population claim but the rigorous validation of a
245 novel paradigm, the identification of dominant confounds, and the specification of a refined
246 theoretical model. In an era in which generative AI is increasingly embedded in children's
247 epistemic environments (Jose, B. et al., 2025; Gerlich, 2025), understanding the conditions under
248 which young children defer to artificial informants is both a developmental and a societal
249 imperative. The present work provides the methodological foundation for that inquiry.

250

251 ***Statements & Declarations***

252 **Funding**

253 This publication is part of CONFIA project PID2024-156166OA-I00, funded by
254 MICIU/AEI/10.13039/501100011033 and by the ESF+; and Consejería de Universidad,
255 Investigación e Innovación, Junta de Andalucía (Grant EMEC_2023_00442).

256 **Competing interests**

257 The author has no relevant financial or non-financial interests to disclose.

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